

INCIDENT RESPONSE REPORT

Ransomware Infection (LockBit) — Workstation Forensic Memory Analysis

Prepared by	Aaron Barrichello Pattinama
Role	SOC Analyst Candidate
Date of Report	14 – August - 2026
Incident Date	13 April 2023
Affected Asset	JOSH-PC (192.168.235.141)
Affected User	Josh (local admin)
Ransomware Family	LockBit
Classification	Confidential — Internal Use Only

1. Executive Summary

Pada 13 April 2023, workstation milik username "Josh" (hostname JOSH-PC, IP 192.168.235.141) terinfeksi ransomware keluarga LockBit. Analisis forensik memori mengonfirmasi eksekusi file berbahaya mal.exe dari direktori Desktop pengguna pada pukul 10:06:45 UTC, yang segera memasang mekanisme persistence melalui Registry Run key, mengenkripsi sedikitnya 1.488 file dengan ekstensi .lockbit, serta melakukan upaya penyebaran lateral (lateral movement) ke lebih dari 200 host lain dalam subnet yang sama melalui port TCP 135 dan 445.

Malware juga terbukti menjalankan rangkaian perintah anti-recovery (penghapusan Volume Shadow Copy dan katalog backup Windows) untuk menghambat proses pemulihan. Snapshot memori diambil saat proses enkripsi masih berlangsung, sekitar 23 detik setelah eksekusi awal, sehingga ransom note belum sempat ditulis ke disk meski identitasnya (Restore-My-Files.txt) telah dikonfirmasi melalui analisis sandbox eksternal terhadap sampel yang identik.

2. Forensic Tools & System Triage

2.1 Forensic Tooling & Methodology

Tools yang digunakan dalam analisis ini:

- Volatility 3 (v2.28.0) — analisis memory forensik dari file evidence.vmem
- VirusTotal — hash lookup, analisis statis, dan analisis dinamis/sandbox untuk identifikasi malware family
- PowerShell (Get-FileHash) — perhitungan hash SHA-256 dari file yang diekstrak

Metodologi / urutan langkah analisis:

- windows.info — identifikasi profil sistem operasi dan waktu snapshot memori
- windows.sessions, windows.getsids, windows.cmdline — identifikasi user context dan proses yang berjalan atas nama masing-masing user
- windows.pslist, windows.pstree, windows.psscan — triase proses untuk menemukan proses mencurigakan berdasarkan nama, path eksekusi, dan hubungan parent-child
- windows.dumpfiles --pid <PID> — ekstraksi executable proses mencurigakan dari memori, dilanjutkan dengan perhitungan hash SHA-256 dan pencarian di VirusTotal
- windows.registry.printkey --key "...CurrentVersion\Run" — pemeriksaan mekanisme persistence pada registry
- windows.filescan — pemindaian seluruh referensi file di memori untuk menemukan artefak dampak ransomware (file terenkripsi, ransom note)
- windows.netscan — pemeriksaan koneksi jaringan aktif untuk mendeteksi indikasi C2 atau lateral movement

```
PS C:\Users\aaaron\Documents\daftar game\privy> vol -f "C:\Users\aaaron\Documents\daftar game\privy\memdump\memdump.vmem" windows.info
Volatility 3 Framework 2.28.0
WARNING volatility3.framework.layers.vmem: No metadata file found alongside VMEM file. A VMSS or VMSN file may be required to correctly process a VMEM file. These should be placed in the same directory with the same file name, e.g. memdump.vmem and memdump.vmss.
Progress: 100.00 PDB scanning finished
Variable Value
Kernel Base 0xf80002a18000
DTB 0xd81000
Symbols file:///C:/Users/aaaron/AppData/Local/Python/pythoncore-3.14-64/Lib/site-packages/volatility3/symbols/windows/ntkrnlmp.pdb/25F4F1A66D0C4385902C9414E61870A2-1.json.xz
Is64Bit True
IsPAE False
layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdDebuggerDataBlock 0xf80002bfb130
NTBuildLab 7601.26111.amd64fre.win7sp1_ldr_
CSDVersion 1
KdVersionBlock 0xf80002bfb0e8
Major/Minor 15.7601
MachineType 34404
KeNumberProcessors 2
SystemTime 2023-04-13 10:07:08+00:00
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 6
NtMinorVersion 1
PE MajorOperatingSystemVersion 6
PE MinorOperatingSystemVersion 1
PE Machine 34404
PE TimeDateStamp Tue Aug 9 03:04:25 2022
```

2.2 System Profiling — User Context

Hostname	JOSH-PC
IP Address	192.168.235.141
OS Version / Build	Windows 7 SP1 x64 (Build 7601.26111)
Affected Username	Josh (local admin — member of Administrators)
Domain / Workgroup	WORKGROUP
CPU Count	2
Memory Snapshot Time	2023-04-13 10:07:08 UTC

Windows 7 SP1 telah mencapai End-of-Life dan tidak lagi menerima security update dari Microsoft ini merupakan faktor risiko signifikan yang perlu ditindaklanjuti.

```
PS C:\Users\aaaron\Documents\daftar game\privy> vol -f "C:\Users\aaaron\Documents\daftar game\privy\memdump\memdump.vmem" windows.sessions
Volatility 3 Framework 2.28.0
WARNING volatility3.framework.layers.vmem: No metadata file found alongside VMEM file. A VMSS or VMSN file may be required to correctly process a VMEM file. These should be placed in the same directory with the same file name, e.g. memdump.vmem and memdump.vmss.
Progress: 100.00 PDB scanning finished
Session ID Session Type Process ID Process User Name Create Time
N/A - 4 System - 2023-04-13 10:00:51.000000 UTC
N/A - 280 smss.exe - 2023-04-13 10:00:51.000000 UTC
0 - 368 csrss.exe /SYSTEM 2023-04-13 10:00:52.000000 UTC
0 - 420 wininit.exe /SYSTEM 2023-04-13 10:00:52.000000 UTC
0 - 480 services.exe /SYSTEM 2023-04-13 10:00:53.000000 UTC
0 - 504 lsass.exe /SYSTEM 2023-04-13 10:00:53.000000 UTC
0 - 520 lsm.exe /SYSTEM 2023-04-13 10:00:53.000000 UTC
0 - 660 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 736 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 788 svchost.exe NT AUTHORITY\LOCAL SERVICE 2023-04-13 10:00:54.000000 UTC
0 - 868 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 912 svchost.exe NT AUTHORITY\LOCAL SERVICE 2023-04-13 10:00:54.000000 UTC
0 - 948 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 312 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 508 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 1140 spoolsv.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:54.000000 UTC
0 - 1180 svchost.exe NT AUTHORITY\LOCAL SERVICE 2023-04-13 10:00:54.000000 UTC
0 - 1556 svchost.exe NT AUTHORITY\LOCAL SERVICE 2023-04-13 10:00:55.000000 UTC
0 - 1672 VGAuthService.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:55.000000 UTC
0 - 1796 vmtoolsd.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:55.000000 UTC
0 - 1824 vmtoolsd.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:55.000000 UTC
0 - 2120 WmiPrivSE.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:55.000000 UTC
0 - 2176 dlhhost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:56.000000 UTC
0 - 2244 msdtc.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:56.000000 UTC
0 - 2648 SearchIndexer.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:01:01.000000 UTC
0 - 2900 WmiPrivSE.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:01:15.000000 UTC
0 - 1336 mscorsvw.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:02:55.000000 UTC
0 - 928 mscorsvw.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:02:56.000000 UTC
0 - 1788 sppsvc.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:02:56.000000 UTC
0 - 2880 SearchProtocolHost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:06:40.000000 UTC
0 - 2516 SearchFilterHost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:06:40.000000 UTC
0 - 612 VSSVC.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:06:45.000000 UTC
0 - 3028 svchost.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:06:45.000000 UTC
0 - 1884 vmtoolsd.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:06:46.000000 UTC
0 - 1944 vds.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:06:46.000000 UTC
1 - 440 csrss.exe /SYSTEM 2023-04-13 10:00:52.000000 UTC
1 - 528 winlogon.exe /SYSTEM 2023-04-13 10:00:53.000000 UTC
1 - 1280 taskhost.exe Josh-PC\Josh 2023-04-13 10:00:54.000000 UTC
1 - 1356 dmw.exe Josh-PC\Josh 2023-04-13 10:00:54.000000 UTC
1 Console 1400 explorer.exe Josh-PC\Josh 2023-04-13 10:00:54.000000 UTC
1 Console 1772 vmtoolsd.exe Josh-PC\Josh 2023-04-13 10:00:55.000000 UTC
1 - 1836 vmtoolsd.exe WORKGROUP\JOSH-PC$ 2023-04-13 10:00:55.000000 UTC
1 - 2392 taskhost.exe Josh-PC\Josh 2023-04-13 10:00:56.000000 UTC
1 - 900 mal.exe Josh-PC\Josh 2023-04-13 10:06:45.000000 UTC
1 - 2812 conhost.exe /SYSTEM 2023-04-13 10:06:45.000000 UTC
1 Console 2460 taskmgr.exe Josh-PC\Josh 2023-04-13 10:06:52.000000 UTC
PS C:\Users\aaaron\Documents\daftar game\privy> vol -f "C:\Users\aaaron\Documents\daftar game\privy\memdump\memdump.vmem" windows.sessions
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3.1 Malware Identification

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Sign in
Sign up

61

Community Score

-40

61/69 security vendors flagged this file as malicious

Reanalyze
Similar
More

5988e75518b2f365671dc49da18b5a70274351721f1f3a8f87bf32984e4024c
file.0xfa801bfe5320.0xfa801c116990.ImageSectionObject.mal.exe.img

Size: 151.50 KB
Last Analysis Date: 22 days ago

peexe persistence detect-debug-environment checks-cpu-name runtime-modules direct-cpu-clock-access long-sleeps overlay calls-wmi

DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY 14+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label
ransomware.lockbit/imps

Threat categories
ransomware trojan

Family labels
lockbitimpsdelshad

Security vendors' analysis

Do you want to automate checks?

AhnLab-V3	Ransomware/Win.LockBit.C3889680	Alibaba	Ransom:Win32/Lockbit.a6aa432f
AllCloud	RansomWare	ALYac	Trojan.Ransom.LockBit
Antiy-AVL	Trojan/Win32.DelShad	Arcabit	Trojan.Ransom.imps.1
Arctic Wolf	Unsafe	Avast	Win32:MalwareX-gen [Ransom]
AVG	Win32:MalwareX-gen [Ransom]	Avira (no cloud)	TR/Crypt.XPACK.Gen7
BitDefender	Gen:Heur.Ransom.Imps.1	Bkav Pro	W32.Malware.EA8CD3EB
ClamAV	Win.Ransomware.Lockbit-10026025-0	CrowdStrike Falcon	Win/malicious_confidence_100% (W)
CTX	Exe.ransomware.lockbit	Cynet	Malicious (score: 99)
DrWeb	Trojan.Encoder.38412	Elastic	Malicious (high Confidence)

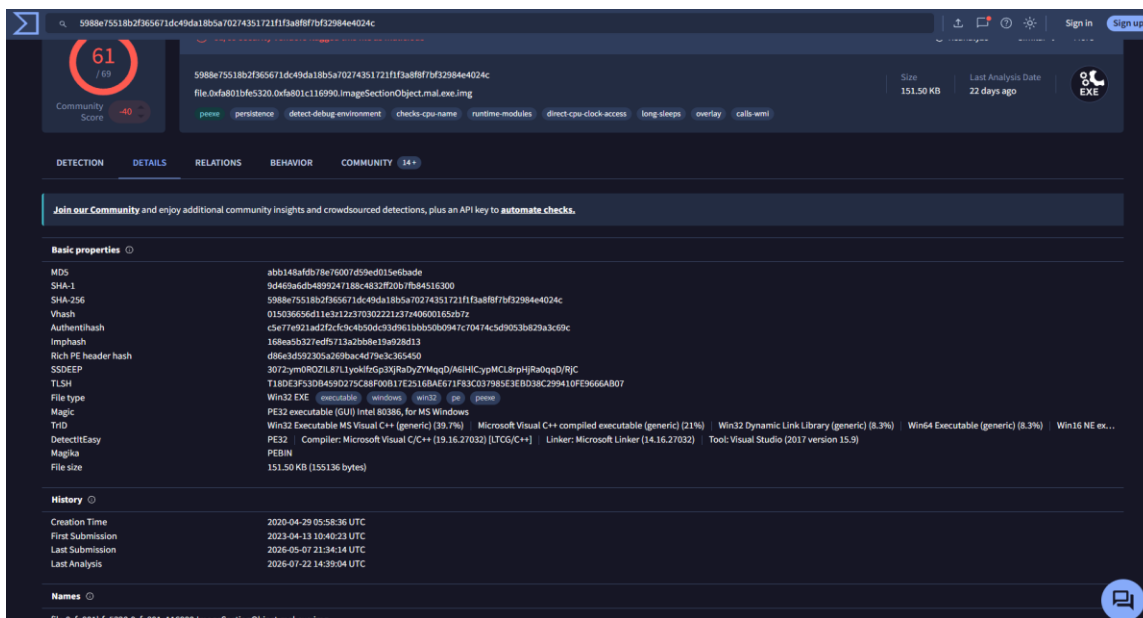
3.2 Malware Integrity — File Hash

File Name	mal.exe (nama asli di riwayat VT: hehehehhe.exe)
SHA-256	5988e75518b2f365671dc49da18b5a70274351721f1f3a8f87bf32984e4024c
MD5	abb148afdb78e76007d59ed015e6bade
File Size	151.50 KB (155,136 bytes)
Compile Timestamp	2020-04-29 05:58:36 UTC

File yang sama dikonfirmasi identik lewat dua region memori berbeda saat proses dumpfiles (ImageSectionObject SHA-256: 5988e755... dan DataSectionObject SHA-256: cea3e8a3...) keduanya dikonfirmasi LockBit oleh puluhan vendor antivirus dengan tingkat kepercayaan 100/100.

```
PS C:\Users\aaaron\Documents\daftar_gawe\privy> Get-FileHash "dump_output\file.0xfa801bfe5320.0xfa801c116990.ImageSectionObject.mal.exe.img" -Algorithm SHA256
```

Algorithm	Hash	Path
SHA256	5988E75518B2F365671DC49DA18B5A70274351721F1F3A8F87BF32984E4024C	C:\Users\aaaron\Documents\daftar_gawe\privy...



3.3 Network Indicators — Lateral Movement

Analisis windows.netscan menemukan pola koneksi yang sangat signifikan: ratusan koneksi TCP dengan status SYN_SENT dari host 192.168.235.141 ke lebih dari 200 alamat IP berbeda dalam subnet /24 yang sama, menargetkan:

- Port TCP 135 (RPC/DCOM) — mayoritas percobaan koneksi
- Port TCP 445 (SMB) — beberapa percobaan, status CLOSED (host tujuan menolak koneksi)

Pola ini konsisten dengan tag perilaku "SPREADER" yang diberikan oleh sandbox VirusTotal, mengindikasikan LockBit mencoba menyebar secara lateral ke mesin lain di jaringan menggunakan protokol SMB/RPC administratif. Tidak ditemukan bukti koneksi C2 eksternal (internet) yang berhasil — seluruh traffic mencurigakan mengarah ke jaringan lokal.

Windows PowerShell									
0x7f403620	TCPv4	192.168.235.141	49585	192.168.235.40	135	SYN_SENT	-	-	N/A
0x7f405010	TCPv4	192.168.235.141	49584	192.168.235.39	135	SYN_SENT	-	-	N/A
0x7f406a70	TCPv4	192.168.235.141	49583	192.168.235.38	135	SYN_SENT	-	-	N/A
0x7f409460	TCPv4	192.168.235.141	49582	192.168.235.36	135	SYN_SENT	-	-	N/A
0x7f40acd0	TCPv4	192.168.235.141	49581	192.168.235.37	135	SYN_SENT	-	-	N/A
0x7f40b260	TCPv4	192.168.235.141	49579	192.168.235.35	135	SYN_SENT	-	-	N/A
0x7f40e730	TCPv4	192.168.235.141	49578	192.168.235.33	135	SYN_SENT	-	-	N/A
0x7f40ecd0	TCPv4	192.168.235.141	49580	192.168.235.34	135	SYN_SENT	-	-	N/A
0x7f4119c0	TCPv4	192.168.235.141	49577	192.168.235.32	135	SYN_SENT	-	-	N/A
0x7f41310	TCPv4	192.168.235.141	49575	192.168.235.31	135	SYN_SENT	-	-	N/A
0x7f415310	TCPv4	192.168.235.141	49574	192.168.235.29	135	SYN_SENT	-	-	N/A
0x7f416010	TCPv4	192.168.235.141	49576	192.168.235.30	135	SYN_SENT	-	-	N/A
0x7f419990	TCPv4	192.168.235.141	49572	192.168.235.28	135	SYN_SENT	-	-	N/A
0x7f419cd0	TCPv4	192.168.235.141	49573	192.168.235.27	135	SYN_SENT	-	-	N/A
0x7f41ea70	TCPv4	192.168.235.141	49571	192.168.235.26	135	SYN_SENT	-	-	N/A
0x7f4203c0	TCPv4	192.168.235.141	49570	192.168.235.24	135	SYN_SENT	-	-	N/A
0x7f421010	TCPv4	192.168.235.141	49569	192.168.235.25	135	SYN_SENT	-	-	N/A
0x7f423460	TCPv4	192.168.235.141	49568	192.168.235.23	135	SYN_SENT	-	-	N/A
0x7f426010	TCPv4	192.168.235.141	49567	192.168.235.22	135	SYN_SENT	-	-	N/A
0x7f426bc0	TCPv4	192.168.235.141	49566	192.168.235.21	135	SYN_SENT	-	-	N/A
0x7f428bc0	TCPv4	192.168.235.141	49565	192.168.235.20	135	SYN_SENT	-	-	N/A
0x7f429310	TCPv4	192.168.235.141	49563	192.168.235.19	135	SYN_SENT	-	-	N/A
0x7f42b2cd	TCPv4	192.168.235.141	49564	192.168.235.18	135	SYN_SENT	-	-	N/A
0x7f42c460	TCPv4	192.168.235.141	49562	192.168.235.17	135	SYN_SENT	-	-	N/A
0x7f42d310	TCPv4	192.168.235.141	49560	192.168.235.16	135	SYN_SENT	-	-	N/A
0x7f431880	TCPv4	192.168.235.141	49561	192.168.235.15	135	SYN_SENT	-	-	N/A
0x7f433460	TCPv4	192.168.235.141	49557	192.168.235.13	135	SYN_SENT	-	-	N/A
0x7f434cd0	TCPv4	192.168.235.141	49559	192.168.235.14	135	SYN_SENT	-	-	N/A
0x7f437310	TCPv4	192.168.235.141	49555	192.168.235.11	135	SYN_SENT	-	-	N/A
0x7f4388c0	TCPv4	192.168.235.141	49551	192.168.235.12	135	SYN_SENT	-	-	N/A
0x7f43a010	TCPv4	192.168.235.141	49553	192.168.235.10	135	SYN_SENT	-	-	N/A
0x7f43b620	TCPv4	192.168.235.141	49548	192.168.235.9	135	SYN_SENT	-	-	N/A
0x7f43d620	TCPv4	192.168.235.141	49546	192.168.235.7	135	SYN_SENT	-	-	N/A
0x7f43e7d0	TCPv4	192.168.235.141	49542	192.168.235.8	135	SYN_SENT	-	-	N/A
0x7f440430	TCPv4	192.168.235.141	49544	192.168.235.6	135	SYN_SENT	-	-	N/A
0x7f444960	TCPv4	192.168.235.141	49540	192.168.235.5	135	SYN_SENT	-	-	N/A
0x7f446010	TCPv4	192.168.235.141	49539	192.168.235.4	135	SYN_SENT	-	-	N/A
0x7f449010	TCPv4	192.168.235.141	49537	192.168.235.3	135	SYN_SENT	-	-	N/A
0x7f449960	TCPv4	192.168.235.141	49522	192.168.235.237	135	SYN_SENT	-	-	N/A
0x7f44b8d0	TCPv4	192.168.235.141	49535	192.168.235.1	135	SYN_SENT	-	-	N/A
0x7f44d010	TCPv4	192.168.235.141	49534	192.168.235.0	135	SYN_SENT	-	-	N/A
0x7f449c10	TCPv4	192.168.235.141	49526	192.168.235.226	135	SYN_SENT	-	-	N/A
0x7f409010	TCPv4	192.168.235.141	49524	192.168.235.226	135	SYN_SENT	-	-	N/A
0x7f40a620	TCPv4	192.168.235.141	49530	192.168.235.217	135	SYN_SENT	-	-	N/A
0x7f40b2cd	TCPv4	192.168.235.141	49532	192.168.235.214	135	SYN_SENT	-	-	N/A
0x7f628cd0	TCPv4	192.168.235.141	49590	192.168.235.44	135	SYN_SENT	-	-	N/A
0x7f6487c0	TCPv4	192.168.235.141	49440	192.168.235.249	135	SYN_SENT	-	-	N/A
0x7f64e7c0	TCPv4	192.168.235.141	49439	192.168.235.248	135	SYN_SENT	-	-	N/A
0x7f654cd0	TCPv4	192.168.235.141	49592	192.168.235.45	135	SYN_SENT	-	-	N/A
0x7f6557c0	TCPv4	192.168.235.141	49502	192.168.235.171	135	SYN_SENT	-	-	N/A
0x7f6577c0	TCPv4	192.168.235.141	49529	192.168.235.220	135	SYN_SENT	-	-	N/A
0x7f659010	TCPv4	192.168.235.141	49533	192.168.235.203	135	SYN_SENT	-	-	N/A
0x7f661cd0	TCPv4	192.168.235.141	49438	192.168.235.245	135	SYN_SENT	-	-	N/A
0x7f6717c0	TCPv4	192.168.235.141	49437	192.168.235.240	135	SYN_SENT	-	-	N/A
0x7f677cd0	TCPv4	192.168.235.141	49435	192.168.235.239	135	SYN_SENT	-	-	N/A
0x7f67b010	TCPv4	192.168.235.141	49436	192.168.235.238	135	SYN_SENT	-	-	N/A
0x7f67ccd0	TCPv4	192.168.235.141	49658	192.168.235.120	135	SYN_SENT	-	-	N/A
0x7f67d7c0	TCPv4	192.168.235.141	49177	192.168.235.236	445	CLOSED	-	-	-
0x7f67fcd0	TCPv4	192.168.235.141	49508	192.168.235.108	135	SYN_SENT	-	-	N/A

4. Persistence, Impact & Timeline Analysis

4.1 Persistence Mechanism

Registry Key Path (exact)	??\C:\Users\Josh\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Run
Value Name	XOIXADp001
Value Data	"C:\Users\Josh\Desktop\mal.exe"
Last Write Time	2023-04-13 10:06:45.000000 UTC

Timestamp modifikasi registry ini identik dengan waktu proses mal.exe pertama kali muncul di process list mengonfirmasi bahwa mekanisme persistence dipasang serentak dengan eksekusi awal malware.

PS C:\Users\aaaron\Documents\daftar gawe\privy> vol -f "C:\Users\aaaron\Documents\daftar gawe\privy\memdump\memdump.vmem" windows.registry.printkey --key "Software\Microsoft\Windows\CurrentVersion\Run"									
Volatility 3 Framework 2.28.0									
WARNING: volatility3.framework.layers.vmware: No metadata file found alongside VMEM file. A VMSS or VMSN file may be required to correctly process a VMEM file. These should be placed in the same directory with the same file name, e.g. memdump.vmem and memdump.vvss.									
Progress: 100.00 PDB scanning finished									
Last Write Time	Hive	Offset	Type	Key	Name	Data	Volatile		
-	0xf8a0000f010	Key	[NONAME]	Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-
-	0xf8a00024010	Key	\REGISTRY\MACHINE\SYSTEM\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
-	0xf8a00059010	Key	\REGISTRY\MACHINE\HARDWARE\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
-	0xf8a00149f010	Key	\SystemRoot\System32\Config\SOFTWARE\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
-	0xf8a00227f010	Key	\SystemRoot\System32\Config\SECURITY\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
-	0xf8a00238a010	Key	\SystemRoot\System32\Config\SAM\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
2023-04-13 10:06:45.000000 UTC	0xf8a00244d010	REG_SZ	??\C:\Users\Josh\ntuser.dat\Software\Microsoft\Windows\CurrentVersion\Run	XOIXADp001	"C:\Users\Josh\Desktop\mal.exe"	False			
2009-07-14 04:45:47.000000 UTC	0xf8a0024d5010	REG_EXPAND_SZ	??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Run	Sidebar %ProgramFiles%\Windows Sidebar\Sidebar.exe /autoRun	False				
2009-07-14 04:45:48.000000 UTC	0xf8a00256a010	REG_EXPAND_SZ	??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Run	Sidebar %ProgramFiles%\Windows Sidebar\Sidebar.exe /autoRun	False				
-	0xf8a0027e6010	Key	??\C:\Users\Josh\AppData\Local\Microsoft\Windows\UsrClass.dat\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
-	0xf8a0030be010	Key	??\C:\System Volume Information\Syscache.hve\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-
-	0xf8a005b32010	Key	\Device\HarddiskVolume1\Boot\BCD\Software\Microsoft\Windows\CurrentVersion\Run	-	-	-	-	-	-

4.2 Ransomware Impact Artifacts

Encrypted File Extension	.lockbit
Jumlah File Terenkripsi (teridentifikasi)	1.488 file (dari windows.filescan)
Ransom Note Filename	Restore-My-Files.txt
Ransom Note — Status di Memory Dump	Tidak ditemukan di snapshot memori ini (lihat Section 4.3)
Ransom Note — Sumber Identifikasi	Analisis sandbox VirusTotal terhadap sampel identik (hash 5988e755...)

```
PS C:\Users\aaroon\Documents\daftar game\privy> Select-String -Path filescan_output3.txt -Pattern "\.lockbit" | Measure-Object -Line
Lines Words Characters Property
-----
1488
```

```
PS C:\Users\aaroon\Documents\daftar game\privy> Select-String -Path filescan_output3.txt -Pattern "\.lockbit" | Select-Object -First 30
filesan_output3.txt:130:0x278e9430 \Program Files\DVD Maker\Shared\DvdStyles\HueCycle\NavigationUp_ButtonGraphic.png.lockbit
filesan_output3.txt:182:0x28bd6950 \Program Files\DVD Maker\Shared\DvdStyles\HueCycle\NavigationRight_ButtonGraphic.png.lockbit
filesan_output3.txt:191:0x28dcff20 \Program Files\DVD Maker\Shared\DvdStyles\Performance\whitemenu.png.lockbit
filesan_output3.txt:193:0x28f645a0 \Program Files (x86)\Windows Sidebar\Gadgets\CPU.Gadget\images\dial.png.lockbit
filesan_output3.txt:197:0x28fb76e0 \Program Files\DVD Maker\Shared\DvdStyles\LayeredTitles\1047x576black.png.lockbit
filesan_output3.txt:199:0x29034ae0 \Program Files\DVD Maker\Shared\DvdStyles\BabyBoy\MainMenuButtonIcon.png.lockbit
filesan_output3.txt:212:0x29549320 \Program Files\DVD Maker\Shared\DvdStyles\BabyBoy\nav_leftarrow.png.lockbit
filesan_output3.txt:234:0x29c40510 \Program Files\DVD Maker\Shared\DvdStyles\Memories\btn-previous-static.png.lockbit
filesan_output3.txt:736:0x7ca04ab0 \Program Files\DVD Maker\Shared\DvdStyles\Full\full.png.lockbit
filesan_output3.txt:749:0x7ca0d750 \Program Files\Windows Sidebar\Gadgets\CPU.Gadget\images\glass_lrg.png.lockbit
filesan_output3.txt:751:0x7ca0da10 \Program Files\Windows Sidebar\Gadgets\PicturePuzzle.Gadget\Images\11.png.lockbit
filesan_output3.txt:843:0x7ca5c070 \Program Files\DVD Maker\Shared\DvdStyles\Full\dotsdarkoverlay.png.lockbit
filesan_output3.txt:889:0x7ca7e3a0 \Program Files\DVD Maker\Shared\DvdStyles\BabyBoy\babyblue.png.lockbit
filesan_output3.txt:916:0x7ca89c80 \Program Files\DVD Maker\Shared\DvdStyles\circlearound_glass.png.lockbit
filesan_output3.txt:923:0x7cafd2e0 \Program Files\DVD Maker\Shared\DvdStyles\Stacking\NavigationUp_SelectionSubpicture.png.lockbit
filesan_output3.txt:967:0x7cb83b70 \Program Files\DVD Maker\Shared\DvdStyles\OldAge\15x15dot.png.lockbit
filesan_output3.txt:968:0x7cb83cc0 \Program Files\DVD Maker\Shared\DvdStyles\HueCycle\NavigationRight_SelectionSubpicture.png.lockbit
filesan_output3.txt:1013:0x7cc64e60 \Program Files\DVD Maker\en-US\OmdProject.dll.mui.lockbit
filesan_output3.txt:1027:0x7cc77f20 \Program Files\VMware\VMware Tools\messages\it\toolboxcmd.vmsg.lockbit
filesan_output3.txt:1032:0x7cc80070 \Program Files\DVD Maker\Shared\DvdStyles\Memories\scrapbook.png.lockbit
filesan_output3.txt:1046:0x7cc91240 \Program Files\Windows Sidebar\Gadgets\Clock.Gadget\icon.png.lockbit
filesan_output3.txt:1049:0x7cc94d20 \Program Files\DVD Maker\Shared\DvdStyles\BabyGirl\flower_trans_rgb.wmv.lockbit
filesan_output3.txt:1063:0x7ccb7910 \Program Files\DVD Maker\Shared\DvdStyles\BabyBoy\BabyBoyMainToNotesBackground_PAL.wmv.lockbit
filesan_output3.txt:1075:0x7cccfd00 \Program Files\DVD Maker\Shared\DvdStyles\Full\NavigationLeft_ButtonGraphic.png.lockbit
filesan_output3.txt:1078:0x7ccd2990 \Program Files (x86)\Windows Sidebar\Gadgets\Weather.Gadget\images\26.png.lockbit
filesan_output3.txt:1080:0x7ccd3070 \Program Files\DVD Maker\Shared\DvdStyles\BabyGirl\16_9-frame-image-mask.png.lockbit
filesan_output3.txt:1083:0x7ccd4f20 \Program Files\DVD Maker\Shared\DvdStyles\HueCycle\15x15dot.png.lockbit
filesan_output3.txt:1111:0x7ccfb800 \Program Files\DVD Maker\Shared\DvdStyles\Push\NavigationLeft_ButtonGraphic.png.lockbit
filesan_output3.txt:1119:0x7cd06700 \Program Files\Windows Sidebar\Gadgets\Calendar.Gadget\drag.png.lockbit
filesan_output3.txt:1126:0x7cd1b710 \Program Files\DVD Maker\Shared\DvdStyles\Memories\16_9-frame-highlight.png.lockbit
```

4.3 Incident Timeline

Kronologi kejadian disusun berdasarkan timestamp yang ditemukan di memory dump:

- 10:00:51–10:02:56 UTC — Proses booting normal sistem Windows (smss, csrss, services, dll)
- 10:06:45 UTC — Proses mal.exe (PID 900) dieksekusi dari C:\Users\Josh\Desktop\; registry Run key XO1XADpO01 dipasang pada detik yang sama (persistence)
- 10:06:45–10:07:08 UTC (±23 detik) — Proses enkripsi file berlangsung aktif; sedikitnya 1.488 file diberi ekstensi .lockbit; ratasan percobaan koneksi lateral movement ke subnet lokal via port 135/445
- 10:07:08 UTC — Snapshot memori diambil; ransom note Restore-My-Files.txt belum sempat ditulis ke disk pada titik ini, mengindikasikan proses enkripsi masih berlangsung aktif saat mesin diisolasi/snapshot diambil

Perintah anti-recovery (vssadmin delete shadows, wmic shadowcopy delete, bcdedit /set recoveryenabled no, wbadmin delete catalog) turut dikonfirmasi lewat analisis sandbox pada sampel identic mengindikasikan Shadow Copy dan katalog backup Windows lokal kemungkinan besar telah dihapus oleh malware sebelum/selama insiden berlangsung.

5. Incident Response Strategy

5.1 Containment

- Putuskan mesin JOSH-PC (192.168.235.141) dari jaringan segera cabut kabel LAN/nonaktifkan adapter WiFi, atau isolasi via VLAN quarantine khusus. Ini prioritas utama mengingat bukti netscan menunjukkan proses aktif mencoba menyebar ke 200+ host lain di subnet yang sama.
- Jangan matikan mesin biarkan tetap menyala (hanya diputus jaringannya) untuk menjaga integritas bukti forensik bila diperlukan analisis lanjutan.
- Blokir IOC di firewall/gateway: hash SHA-256 (5988e755...), domain lockbit-decryptor.com, dan pola koneksi mencurigakan ke port 445/135 antar-host non-server.
- Nonaktifkan akun Josh sementara di Active Directory/local admin untuk mencegah penggunaan kredensial yang sama di mesin lain.
- Scan mesin-mesin lain di subnet 192.168.235.0/24 yang sempat menjadi target percobaan koneksi (SYN_SENT) walau status mayoritas gagal/closed, perlu verifikasi tidak ada mesin lain yang turut terinfeksi.
- Amankan salinan forensik lengkap (disk image, bukan hanya memory dump) sebelum melakukan tindakan lebih lanjut, untuk kebutuhan investigasi dan keperluan legal.

5.2 Eradication

- Hapus file malware: C:\Users\Josh\Desktop\mal.exe (SHA-256: 5988e755...).
- Hapus registry persistence key: HKCU\Software\Microsoft\Windows\CurrentVersion\Run, value XO1XADpO01.
- Jalankan full scan dengan endpoint protection/EDR yang sudah diperbarui signature LockBit terbaru untuk memastikan tidak ada komponen tersisa (dropper lain, scheduled task, dll).
- Periksa Scheduled Tasks secara manual LockBit dikenal juga memanfaatkan Task Scheduler sebagai mekanisme persistence tambahan yang belum terverifikasi dalam analisis ini.
- Reset seluruh kredensial yang berpotensi tersimpan/digunakan di mesin ini (local admin, cached domain credentials), mengingat proses berjalan dengan privilege administrator.
- Pertimbangkan wipe & reimage total mesin daripada cleaning manual untuk ransomware modern seperti LockBit, reimage jauh lebih aman dibanding mengandalkan removal tool karena risiko residual backdoor.

5.3 Recovery & Hardening

- Pulihkan data dari backup offline/immutable. PENTING: LockBit pada kasus ini terbukti menjalankan vssadmin delete shadows /all, wmic shadowcopy delete, dan wbadmin delete catalog Shadow Copy dan Windows Backup lokal kemungkinan besar sudah terhapus. Jangan andalkan System Restore; gunakan backup eksternal/cloud yang terpisah dari jaringan yang terinfeksi.
- Verifikasi integritas backup sebelum restore pastikan backup yang dipakai diambil sebelum tanggal insiden (13 April 2023) dan bebas dari malware.
- Reimage mesin JOSH-PC dari base image bersih, bukan sekadar recovery di tempat.
- Patch & update: OS Windows 7 SP1 sudah end-of-life dan sangat rentan migrasikan ke versi yang masih didukung, atau minimal terapkan seluruh patch keamanan yang tersedia bila migrasi belum memungkinkan.
- Harden SMB/RPC: nonaktifkan SMBv1, batasi port 135/445 hanya untuk komunikasi yang benar-benar diperlukan (segmentasi jaringan/firewall internal) ini jalur utama yang dieksploitasi untuk lateral movement pada insiden ini.
- Terapkan application whitelisting (mis. AppLocker/WDAC) untuk mencegah eksekusi executable dari folder Desktop/Downloads/Temp, mengingat mal.exe dijalankan langsung dari Desktop user.

- Terapkan prinsip least privilege cabut hak local administrator dari akun Josh jika tidak esensial untuk pekerjaannya; proses malware berjalan dengan privilege admin yang memperbesar dampak, termasuk kemampuan menghapus shadow copy.

5.4 Post-Incident Recommendation

- Terapkan strategi backup 3-2-1 dengan salinan offline/air-gapped atau immutable (mis. cloud backup dengan object lock) agar ransomware yang menghapus shadow copy lokal tidak dapat menyentuh backup tersebut.
- Terapkan segmentasi jaringan (network segmentation) untuk membatasi broadcast domain, sehingga satu mesin terinfeksi tidak dapat langsung menjangkau ratusan host lain seperti yang terjadi pada insiden ini.
- Tingkatkan security awareness training bagi pengguna titik masuk awal (mal.exe di Desktop) mengindikasikan kemungkinan unduhan/eksekusi manual, sehingga edukasi terkait phishing dan verifikasi file sebelum eksekusi perlu ditingkatkan.
- Deploy EDR/XDR dengan kemampuan behavioral detection (bukan hanya signature-based AV) — pola LockBit di sini mencakup penghapusan shadow copy dan modifikasi bcdedit yang dapat dideteksi lewat behavior rule (mis. Sigma rule "Delete shadow copy via WMIC", "CurrentVersion Autorun Keys Modification" yang sudah match di VirusTotal).
- Perbarui playbook Incident Response khusus ransomware, mencakup prosedur isolasi cepat berbasis network segmentation otomatis (mis. NAC yang dapat auto-isolate host begitu mendeteksi mass-scanning ke port 135/445).
- Audit dan nonaktifkan protokol legacy (SMBv1, RPC unrestricted) di seluruh jaringan sebagai kebijakan baseline, bukan hanya di mesin yang terdampak.
- Perpanjang retensi log jaringan (NetFlow/firewall) untuk mendukung investigasi forensik retroaktif pada kasus ini, konfirmasi apakah mesin lain benar-benar berhasil dieksploitasi memerlukan log jaringan yang lebih luas dari sekadar memory dump satu mesin.

6. Indicators of Compromise (IOC) Summary

Type	Value
Malware Filename	mal.exe
Malware SHA-256	5988e75518b2f365671dc49da18b5a70274351721f1f3a8f8f7bf32984e4024c
Malware MD5	abb148afdb78e76007d59ed015e6bade
Ransomware Family	LockBit
Encrypted File Extension	.lockbit
Ransom Note Filename	Restore-My-Files.txt
Persistence Registry Key	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\XO1XADpO01
Lateral Movement Ports	TCP 135 (RPC), TCP 445 (SMB)
Related Domain (dari sandbox)	lockbit-decryptor.com
Affected Host	JOSH-PC (192.168.235.141)